

- Description

This module manages a smart gym equipment usage monitoring and load balancing system, ensuring that:

1. **Gym equipment usage is continuously monitored to prevent overcrowding at specific machines.**
2. **If certain machines are overutilized, members are suggested alternative exercises or wait times are optimized.**
3. **If certain machines are underutilized, members are encouraged to use them to balance the load.**
4. **A log of all equipment usage, alternative exercise recommendations, and gym occupancy levels is maintained for better fitness management.**

This system helps improve gym efficiency, reduce waiting times, and optimize equipment distribution.

- Specification

module SmartGymEquipmentUsage

var currentUsers, maxCapacity: integer

var log: sequence of string

process useEquipment(numUsers: integer)

functional scenario form:

currentUsers + numUsers <= maxCapacity && currentUsers = currentUsers + numUsers, log = log $\oplus$ ("Members Started Using Equipment | New Count: " + currentUsers + "/" + maxCapacity), Members successfully started using the equipment.

||

currentUsers + numUsers > maxCapacity && log = log $\oplus$ ("Overcrowding Alert: Suggested alternative exercises."), Overcrowding warning triggered, suggesting alternatives.

end process

process leaveEquipment(numUsers: integer)

functional scenario form:

Members left the equipment, usage updated.

end process

end module

- Java Code

```
package code;
import java.util.*;
class GymEquipment {
    private String name;
    private int maxCapacity;
    private int currentUsers;
    private List<String> log;
```

```

    public GymEquipment(String name, int maxCapacity) {
        this.name = name;
        this.maxCapacity = maxCapacity;
        this.currentUsers = 0;
        this.log = new ArrayList<>();
    }

    public void useEquipment(int numUsers) {
        if (currentUsers + numUsers > maxCapacity) {
            System.out.printf("WARNING: %s is Overcrowded! Suggesting A  
lternative Exercises.%n", name);
            log.add("Overcrowding Alert: Suggested alternative exercise  
s.");
            return;
        }
        currentUsers += numUsers;
        System.out.printf("%d Members Started Using %s | Current Users:  
%/d/%d%n", numUsers, name, currentUsers, maxCapacity);
        log.add(String.format("%d Members Started Using | New Count: %d  
/%d", numUsers, currentUsers, maxCapacity));
    }

    public void leaveEquipment(int numUsers) {
        currentUsers = Math.max(0, currentUsers - numUsers);
        System.out.printf("%d Members Left %s | Remaining Users: %d/%d%  
n", numUsers, name, currentUsers, maxCapacity);
        log.add(String.format("%d Members Left | Remaining Count: %d/%d  
", numUsers, currentUsers, maxCapacity));
    }

    public void printLog() {
        System.out.printf("\n%s Usage Log:%n", name);
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c58_SmartGymSystem {
    public static void main(String[] args) {
        GymEquipment treadmill = new GymEquipment("Treadmill", 5);
        GymEquipment benchPress = new GymEquipment("Bench Press", 3);
        // testcase-VT:
        //     treadmill.useEquipment(3);
        //     treadmill.useEquipment(2);
        //     treadmill.useEquipment(1);
        //     treadmill.leaveEquipment(2);
        //     benchPress.useEquipment(2);
    }
}

```

```

//      benchPress.leaveEquipment(1);
//      treadmill.printLog();
//      benchPress.printLog();
//      // testcase-FT:
//      treadmill.useEquipment(-574478552);
//      treadmill.useEquipment(-656877352);
//      treadmill.useEquipment(-29);
//      treadmill.useEquipment(0);
//      benchPress.useEquipment(-61);
//      benchPress.useEquipment(0);
//      treadmill.printLog();
//      benchPress.printLog();

//      // testcase-Z3:
//      treadmill.useEquipment(3);
//      treadmill.useEquipment(2);
//      treadmill.useEquipment(1);
//      treadmill.leaveEquipment(2);
//      benchPress.useEquipment(2);
//      benchPress.leaveEquipment(1);
//      treadmill.printLog();
//      benchPress.printLog();

//      // testcase-UVT:
//      treadmill.useEquipment(3);
//      treadmill.useEquipment(2);
//      treadmill.useEquipment(1);
//      treadmill.leaveEquipment(2);
//      benchPress.useEquipment(2);
//      benchPress.leaveEquipment(1);
//      treadmill.printLog();
//      benchPress.printLog();
    }
}

```

- Coverage report

1. Test case number: 6
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c58_SmartGymSystem.java	100.0 %	4	0	4
> c58_SmartGymSystem		0	0	0
> GymEquipment	100.0 %	4	0	4

3. Vibration testing instruction coverage: 98.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	225	20,417	20,642
▼ src	1.1 %	225	20,417	20,642
▼ code	1.1 %	225	20,417	20,642
▼ c58_SmartGymSystem.java	98.7 %	225	3	228
> c58_SmartGymSystem	92.1 %	35	3	38
> GymEquipment	100.0 %	190	0	190

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	839	842
▼ src	0.4 %	3	839	842
▼ code	0.4 %	3	839	842
▼ c58_SmartGymSystem.java	75.0 %	3	1	4
> c58_SmartGymSystem		0	0	0
> GymEquipment	75.0 %	3	1	4

5. Fuzz testing instruction coverage: 64%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	146	20,496	20,642
▼ src	0.7 %	146	20,496	20,642
▼ code	0.7 %	146	20,496	20,642
▼ c58_SmartGymSystem.java	64.0 %	146	82	228
> c58_SmartGymSystem	92.1 %	35	3	38
> GymEquipment	58.4 %	111	79	190

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c58_SmartGymSystem.java	100.0 %	4	0	4
> c58_SmartGymSystem		0	0	0
> GymEquipment	100.0 %	4	0	4

7. Z3 instruction coverage: 98.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	225	20,417	20,642
▼ src	1.1 %	225	20,417	20,642
▼ code	1.1 %	225	20,417	20,642
▼ c58_SmartGymSystem.java	98.7 %	225	3	228
> c58_SmartGymSystem	92.1 %	35	3	38
> GymEquipment	100.0 %	190	0	190

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c58_SmartGymSystem.java	100.0 %	4	0	4
> c58_SmartGymSystem		0	0	0
> GymEquipment	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 98.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	225	20,417	20,642
▼ src	1.1 %	225	20,417	20,642
▼ code	1.1 %	225	20,417	20,642
▼ c58_SmartGymSystem.java	98.7 %	225	3	228
> c58_SmartGymSystem	92.1 %	35	3	38
> GymEquipment	100.0 %	190	0	190